∂T

Partion Function of single 1D quantum harmonic oscillator in High T limit.

$$Z = \frac{K_B T}{\hbar \omega}$$, using units such that $K_B = 1$, $\hbar = 1$

$$Z = \frac{T}{\omega}$$

For N 1D harmonic oscillators

$$Z_N = Z^N$$

$$\therefore Z_N = \left(\frac{T}{\omega}\right)^N$$

$$E = T^2 N \frac{\partial \ln Z}{\partial T} = T^2 N \frac{\partial}{\partial T}(\ln T - \ln \omega)$$

$$= N T^2 \frac{1}{T} = NT$$

$$\therefore E = NT$$

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